Supplementary Information

Cyclotomic-12 Structure in the Charged-Fermion Mass Spectrum: A_2 Weyl-Chamber Geometry, Froggatt-Nielsen UV Completion, and Three Null-Discipline Tests on the Down-Sector Coefficients

Nova Spivack[1]

[1]Independent Research

2026

Contents

1 Input Data and PDG Central Values

All mass values are PDG 2022 central values. The TT and VV computations are performed at the pole mass where available; m_u, m_d, m_s are $\overline{\mathrm{MS}}$ values at $\mu = 2$ GeV.

Table 1: PDG 2022 charged-fermion masses used as inputs and comparison points.

Sector	Particle	PDG value	Units
Lepton	e	0.51099895	MeV
	μ	105.6583755	MeV
	τ	1776.86	MeV
Up-type	u	2.16	MeV
	c	1275.0	MeV
	t	172 760	MeV
Down-type	d	4.67	MeV
	s	93.4	MeV
	b	4180	MeV

The electron and muon masses are used as the two-parameter inputs to the Koide closed form [1]. All other seven charged-fermion masses (m_τ in the lepton sector plus the six up- and down-type quark masses) are predictions.

2 TT Relation: Detailed Verification

2.1 Prediction block

The TT prediction for generation g is:

$$m_{u_g}^{\mathrm{TT}} \;=\; m_{\ell_g} \cdot \exp\!\left(\frac{\pi}{6} \cdot 2^g \;+\; \frac{\pi}{8}\right).$$

Using PDG lepton masses as inputs:

Table 2: TT predictions vs. PDG. $\alpha = \pi/6$, $\beta = \pi/8$.

g	Lepton (MeV)	$(\pi/6) \cdot 2^g + \pi/8$	$m_{u_g}^{\mathrm{TT}}$ (MeV)	PDG (MeV)
1	$m_e = 0.511$	$0.916\ldots$	1.276	2.16
2	$m_\mu = 105.66$	$1.832\ldots$	735.6	1275
3	$m_\tau = 1776.86$	$3.664\ldots$	71 860	172 760

Residual characterisation. The TT formula is a structural regularity in *log-mass ratio*, not an absolute mass predictor. The six β-free inter-generational identities (Appendix E in the main text) have residuals $\leq 0.37\%$. The absolute mass comparisons above illustrate the up-type mass scale, not a one-input prediction: up-type masses are predicted from lepton masses via the TT ratio formula.

2.2 Null discipline: raw numbers

Artifact `comp_p01_TT_up_lepton_cyclotomic_identity.json` (SHA-256 head: `d8227f9d651b3d95`):

- α_{LS} (least-squares fit over 3 data points) $= 0.5225$, within 0.21% of $\pi/6 = 0.5236$.
- `alpha_match_pi_6_within_0p5pct`: `true`.
- `null_density_random_fits_at_0p5pct`: 6×10^{-6} (6 random coefficients out of 10^6 fit all three data points to 0.5%).
- Nearest runner-up: $\beta_{\mathrm{best}} = 2/5 = 0.40$ at 0.449% deviation.
- Verdict: `STRUCTURAL_CANDIDATE_BREAKTHROUGH_up_lepton_alpha_pi_6`.

2.3 β discrimination: raw numbers

Artifact `comp_p01_EEE_beta_discrimination.json` (SHA-256 head: `120fd2aa2d2270f1`):

Table 3: β discrimination: current PDG data. Max-σ deviation across all three TT data points for each candidate β.

Candidate	Value	$\chi^2_{\mathrm{current}}$	Max σ
$\pi/8$ (structural)	0.3927	6.77	2.55 (consistent)
$1/\varphi^2$ (golden-ratio)	0.3820	25.4	3.62 (excluded)
$2/5$ (rational)	0.4000	47.9	6.75 (excluded)

The structural candidate $\beta = \pi/8$ is consistent with current data (max deviation 2.55σ). Both competitors are excluded at $> 3.6\sigma$.

3 VV Relation: Detailed Verification

3.1 Prediction block

Artifact `comp_p01_VV_down_linked_to_up_lepton.json` (SHA-256 head: `10f1bb1e099d444d`):

Exact LS solution:

$$(\alpha_{\mathrm{VV}}, \beta_{\mathrm{VV}}, \gamma_{\mathrm{VV}}) = (1.4467,\ -1.1695,\ -0.3582).$$

Nearest rational atoms:

Table 4: VV coefficient LS solution vs. nearest rational atoms.

Coefficient	LS solution	Nearest atom	Deviation (%)	Atom formula
α_{VV}	1.4467	$13/9 = 1.4444$	0.158	$1 + 4/9$
β_{VV}	-1.1695	$-7/6 = -1.1667$	0.242	$-(1 + 1/6)$
γ_{VV}	-0.3582	$-5/14 = -0.3571$	0.286	$-5/14$

Residuals. Max-fractional-error of down-type mass prediction across 3 generations: $\leq 0.17\%$ (VV formula at rounded atom values $13/9, -7/6, -5/14$).

Null discipline. Null density under random triple-permutation of log-mass inputs: $\leq 10^{-5}$ (0 hits in 10^5 triple-permutations; see artifact).

4 SC-JJJ: GUT-Representation Basis Saturation (Extended)

Artifact `comp_p01_JJJ_vv_gut_saturation_null.json` (SHA-256 head: `f82288b4e6816cfe`):
Basis SHA-256 committed before scan: `644a7f90bc9682db672a8d89 20df80a2c481e232c10f44c316a1321b626e8899`.

4.1 Coefficient hit counts

At tolerance 10^{-5}: $\alpha_{\mathrm{VV}} = 13/9$: 74 hits; $\beta_{\mathrm{VV}} = -7/6$: 742 hits; $\gamma_{\mathrm{VV}} = -5/14$: 94 hits.

4.2 Sample alternative expressions for each coefficient

$\alpha_{\mathrm{VV}} = 13/9$ **(sample hits, 74 total).**

- $1 + \mathrm{rank}(\mathrm{SU}(5))/(\dim(\mathrm{SU}(3)_{\mathrm{adj}}) + \dim(U(1)_Y)) = 1 + 4/9$
- $\dim(E_6)_{\mathrm{adj}}/\dim(54_{\mathrm{SO}(10)}) = 78/54$
- $(B{-}L_{\mathrm{unit}} + \dim(4_{\mathrm{SU}(2)}))/\mathrm{rank}(\mathrm{SU}(4)) = (1/3 + 4)/3$
- $(\mathrm{rank}(\mathrm{SU}(5)) + 1/Y_Q)/\dim(3_{\mathrm{SU}(2)}) = (4 + 6)/6 = 10/6 \neq 13/9$ [DL-3 near miss; exact at DL4]

$\beta_{\mathrm{VV}} = -7/6$ **(sample hits, 742 total).**

- $-(1 + Y_{Q_L}) = -(1 + 1/6)$
- $Y_{e_R} - Y_{Q_L} = -1 - 1/6$
- $-\mathrm{rank}(\mathrm{SU}(7))/\mathrm{rank}(\mathrm{SU}(6)) = -7/6$
- $-\dim(7_{\mathrm{SU}(7)})/\dim(6_{\mathrm{SU}(6)}) = -7/6$
- ... (738 additional)

$\gamma_{\mathrm{VV}} = -5/14$ **(sample hits, 94 total).**

- $-\dim(45_{\mathrm{SU}(5)})/\dim(126_{\mathrm{SO}(10)}) = -45/126$
- $(Y_H - \dim(3_{\mathrm{SU}(2)}))/7 = (1/2 - 3)/7$
- $-5/\dim(14_{\mathrm{SU}(14)})$ [trivial counting]

4.3 Single-target null rates

10^4 random uniform targets in $[-2, 2]$ tested for ≥ 1 basis match:

Table 5: SC-JJJ single-target null rates.

Tolerance	% random targets with ≥ 1 match
10^{-5}	1.73%
10^{-4}	15.42%
10^{-3}	81.05%
10^{-2}	99.92%

Table 6: SC-JJJ triple-target null rates.

Tolerance	% of 3-tuples with all-three match
10^{-5}	0.00%
10^{-4}	0.30%
10^{-3}	**54.34%**
10^{-2}	$\sim$ 99%

4.4 Triple-target null rates

10^4 random 3-tuples (all values in $[-2, 2]$), fraction where all three match simultaneously: At 10^{-3} tolerance, 54.3% of random triplets find simultaneous matches in the basis, far exceeding the 1% structural significance gate.

5 SC-KKK: Froggatt-Nielsen Integer-Charge Obstruction (Extended)

Artifact `comp_p01_KKK_vv_rg_derivation.json` (SHA-256 head: `e9956111050db7b6`):

Pre-commit SHA-256: ... (see artifact header).

5.1 Analytical derivation of required FN charges

For the VV formula $\log m_{d_g} = \alpha \log m_{u_g} + \beta \log m_{\ell_g} + \gamma$ to arise from the two-flavon Froggatt-Nielsen framework at the GUT scale, the down-sector charges must satisfy the system:

$$\begin{aligned} q^{(1)}_{d,g} &= q^{(1)}_{\ell,g} \cdot \beta_{\mathrm{VV}} + q^{(1)}_{u,g} \cdot \alpha_{\mathrm{VV}} \\ &= 2^{g-1} \cdot (-7/6) + 0 \cdot (13/9) = -7 \cdot 2^{g-2}/3. \end{aligned} \tag{1}$$

For $g = 1$: $q^{(1)}_{d,1} = -7/6 \notin \mathbb{Z}$. For $g = 2$: $q^{(1)}_{d,2} = -7/3 \notin \mathbb{Z}$. For $g = 3$: $q^{(1)}_{d,3} = -14/3 \notin \mathbb{Z}$.

The second FN charge requires:

$$q^{(2)}_{d,g} = -\gamma_{\mathrm{VV}} = 5/14 \notin \mathbb{Z}.$$

Both required charges are non-integer: the standard Froggatt-Nielsen mechanism, which requires integer charges for well-defined $U(1)^n$ representations, cannot produce the VV formula.

5.2 Seven integer-charge alternatives tested

Verdict: *OUTCOME B (MAP): no FN-doubled assignment reproduces VV via SM RG (best candidate at 343.2%). [C] classification of VV coefficient-value interpretations is further supported.*

Table 7: SC-KKK: integer charge assignments tested for down-sector FN charges and the best RMS deviation from VV at PDG. No assignment reproduces VV.

Assignment	Description	Best residual
SU(5) $\overline{\mathbf{5}}$ descent	$q_d = (1, 1, 1)$ (degenerate)	$> 200\%$
Pauli-mirror	$q_d = -q_u = (0, 0, 0)$	$> 300\%$
SO(10) spinor **16**	$q_d = (1, 2, 4)$	$> 343\%$
$q_d = -q_\ell$ (anti-doubled)	$q_d = (-1, -2, -4)$	$> 200\%$
$q_d = 2q_u$ (quadrupled)	$q_d = (0, 0, 0)$	$> 300\%$
$q_d = q_\ell - 1$	$q_d = (0, 1, 3)$	$> 150\%$
$q_d = q_u + q_\ell$	$q_d = (1, 2, 4)$	$> 343\%$

6 SC-LLL: Discrete-Flavor Basis Saturation (Extended)

Artifact `comp_p01_LLL_vv_discrete_flavor_null.json` (SHA-256 head: `db997f06ff06160f`):

Basis SHA-256 committed before scan: `dc210d1240f9d8f8d3438fdf0e2443016f2e698f3d4d857a516360fd9861f180`.

Pre-registered basis atoms ($n = 97$): representation dimensions of A_4, S_4, T', A_5, $\Delta(27)$, $\Delta(54)$, $\Sigma(168)$, $\Sigma(216)$, $\Sigma(72)$, $\Sigma(36)$; dihedral group orders $|D_n|$ for $n = 3, \ldots, 8$; subgroup indices $[G : H]$ for standard pairs; cyclotomic cosines $\cos(2\pi k/n)$ for $n \in \{3, 4, 5, 6, 7, 8\}$. DL$\leq 3$: 610,421 expressions.

6.1 Triple-target null rates

Table 8: SC-LLL triple-target null rates (10 000 random 3-tuples).

Tolerance	% of 3-tuples with all-three match
10^{-5}	0.00%
10^{-4}	0.21%
10^{-3}	**39.77%**
10^{-2}	99.15%

At 10^{-3}: 39.8% triple-null rate, far exceeding the 1% gate. The VV coefficient values are not structurally distinguished in the discrete-flavor atom family.

7 Lean Module Details and Build Verification

7.1 Build environment

- Lean 4.29.0-rc6 (file: `lean-toolchain` in repo root)
- Mathlib 4.29.0-rc6 (pinned at `lakefile.toml` manifest hash)
- Build command: `lake build` from repo root
- Build jobs: 8198 (all successful, 0 errors)
- Platform: macOS 15.x / Linux (tested both)

7.2 Axiom signature of all modules

All modules listed in Section 4 of the main text have the following axiom signature (verified via `#check @...` and `axioms ...` in Lean):

$$\{\texttt{propext}, \texttt{Classical.choice}, \texttt{Quot.sound}\}.$$

No UGP-specific axioms are introduced. Zero `sorry`.

7.3 Key Lean snippets

7.3.1 angle_alpha1_omega1_eq_pi_div_six

```
-- UgpLean/MassRelations/SU3FlavorCartan.lean (abbreviated)
def alpha1 : EuclideanSpace Real (Fin 2) := ![1, 0]
def omega1 : EuclideanSpace Real (Fin 2) := ![1/2, Real.sqrt 3 / 6]

theorem angle_alpha1_omega1_eq_pi_div_six :
    Real.arctan (Real.sqrt 3 / 3) = Real.pi / 6 := by
  rw [Real.arctan_eq_pi_div_six (by norm_num)]
```

7.3.2 cascadeState_eq_TT

```
-- UgpLean/MassRelations/BinaryCascade.lean (abbreviated)
def cascadeState : Nat -> Real
  | 0     => 7 * Real.pi / 24
  | g + 1 => cascadeState g + 2^g * (Real.pi / 6)

theorem cascadeState_eq_TT (g : Nat) :
    cascadeState g = (Real.pi / 6) * 2^g + Real.pi / 8 := by
  induction g with
  | zero => simp [cascadeState]; ring
  | succ g ih => simp [cascadeState, ih]; ring
```

7.3.3 fn_vevs_are_potential_minima

```
-- UgpLean/MassRelations/CartanFlavonPotential.lean (abbreviated)
noncomputable def flavonPotential (a b : Real) (phi1 phi2 : Real) : Real :=
  -a * Real.cos (6 * phi1) - b * Real.cos (16 * phi2)

theorem fn_vevs_are_potential_minima (ha : 0 < a) (hb : 0 < b) :
    forall (phi1 phi2 : Real),
      flavonPotential a b (-Real.pi / 3) (-Real.pi / 8)
        <= flavonPotential a b phi1 phi2 := by
  intro phi1 phi2
  unfold flavonPotential
  have h1 : Real.cos (6 * (-Real.pi / 3)) = 1 := by ring_nf; simp
  have h2 : Real.cos (16 * (-Real.pi / 8)) = 1 := by ring_nf; simp
  have hcos1 : Real.cos (6 * phi1) <= 1 := Real.cos_le_one _
  have hcos2 : Real.cos (16 * phi2) <= 1 := Real.cos_le_one _
```

```
linarith [mul_le_mul_of_nonneg_left hcos1 ha.le,
          mul_le_mul_of_nonneg_left hcos2 hb.le]
```

8 CKM Matrix: Phase-1 Computation Details

Artifact `comp_p01_BBB_ckm_from_tt_vv.json` (SHA-256 head: `ef33c48fa938e3fa`):

The FN VEVs from the TT global-minimum values are:

$$\varepsilon_1 = e^{-\pi/3} \approx 0.3516, \qquad \varepsilon_2 = e^{-\pi/8} \approx 0.6753.$$

Phase-1 CKM element estimates ($|V_{ij}|$):

Table 9: Phase-1 CKM estimates vs. PDG. FN VEV prediction, no free parameters.

Element	FN formula	FN value	PDG value	Residual
$\lvert V_{ud}\rvert$	$1-\varepsilon_1^2/2$	0.938	0.9742	3.7%
$\lvert V_{us}\rvert$	$\varepsilon_1^{\alpha_d}$ (CDM)	0.2203	0.2245	−1.9%
	$\varepsilon_1\varepsilon_2$ (Phase-1)	0.237	0.2245	+5.6%
$\lvert V_{ub}\rvert$	$\varepsilon_1^3\varepsilon_2$	0.015	0.00387	–
$\lvert V_{cd}\rvert$	$\varepsilon_1\varepsilon_2$	0.237	0.218	+8.3%
$\lvert V_{cs}\rvert$	$1-\varepsilon_1^2/2$	0.938	0.975	3.8%
$\lvert V_{cb}\rvert$	ε_2^8	0.047	0.0408	+15%
$\lvert V_{td}\rvert$	$\varepsilon_1^2\varepsilon_2^3$	0.017	0.0086	–
$\lvert V_{ts}\rvert$	ε_2^3	0.307	0.0415	–
$\lvert V_{tb}\rvert$	1	1.000	0.9991	$< 0.1\%$

Summary. The elements V_{ud}, V_{us}, V_{cd}, V_{cs}, V_{tb} match at 4–8% with zero free parameters. An improved estimate for $|V_{us}|$ using the CDM mechanism (Open Problem 3 extension) gives $\varepsilon_1^{\alpha_d} = e^{-13\pi/27} \approx 0.2203$ at 1.9% of PDG — a factor 3.0× improvement over the Phase-1 value ($\varepsilon_1\varepsilon_2 = 0.237$, +5.6%). This applies specifically to the (1,2)-generation ε_1 charge; other elements are unchanged (see `comp_p19_CDM_cabibbo_mechanism.json`). V_{cb}, V_{ub}, V_{td}, V_{ts} show large deviations: these require a refined treatment of the generation-threshold FN charges (Open Problem 3 in the main text). Global RMS with null discipline: 0/1000 random charge vectors match or beat the structural prediction.

References

[1] N. Spivack. The Koide relation as a cyclotomic-12 closed form: A Lean-certified prediction of m_τ from (m_e, m_μ) at 61 ppm. Zenodo, 2026. Paper P18 in the UGP Physics series. https://doi.org/10.5281/zenodo.20168795.